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A Correlating Equation for Forced Convection From Gases and Liquids to a Circular Cylinder in Crossflow

A single comprehensive equation is developed for the rate of heat and mass transfer from a circular cylinder in crossflow, covering a complete range of Pr (or Sc) and the entire range of Re for which data are available. This expression is a lower bound (except possibly for RePr < 0.2); free-stream turbulence, end effects, channel blockage, free convection, etc., may increase the rate. In the complete absence of free convection, the theoretical expression of Nakai and Okazaki may be more accurate for RePr < 0.2. The correlating equation is based on theoretical results for the effect of Pr in the laminar boundary layer, and on both theoretical and experimental results for the effect of Re. The process of correlation reveals the need for theoretical results for the effect of Pr in the region of the wake. Additional experimental data for the effect of Pr at small Pe and for the effect of Re during the transition in the point of separation are also needed.

Introduction

Convective heat transfer from a cylinder in crossflow has been the subject of repeated attempts at correlation and generalization because of its importance in a variety of applications. These attempts have not been completely successful because of: (1) the lack of a comprehensive theory for the dependence on Pr even for the boundary layer regime; (2) competitive theories for low Re; (3) the influence of natural convection at very low Re; (4) discrete transitions in the boundary layer and the wake at high Re; (5) the influence and lack of specification of free-stream turbulence; (6) the use of different and undefined thermal boundary conditions; (7) significant variation in physical properties between the surface and free stream and around the surface; (8) the incorporation of erroneous physical properties in older tabulated data; (9) end effects, particularly at low Re; (10) tunnel blockage; (11) significant scatter in most of the many sets of data; and (12) unresolved discrepancies between the various sets of data.

Douglas and Churchill [1][2] discovered that most of the experimental data for gases at large temperature differences had been misinterpreted and misplotted in McAdams [2]. They reconstructed a general graphical correlation for gases using the original measured quantities and modern physical properties. Fand and Keswani [3] recently repeated the conversion of Hilpert's [4] extensive data for air to modern physical properties.

Douglas and Churchill also proposed a semitheoretical equation for all Re and attributed the systematic deviation at low Re to free convection. (This behavior has since been shown to be due to thickening of the boundary layer, as well as to free convection.) Perkins and Leppert [5] developed graphical correlations and empirical equations for liquids based on the models of Richardson [6] and of Douglas and Churchill. Tsubouchi and Masuda [7] carried out a thorough examination of the data for both liquids and gases. They proposed a new equation for liquids and a slight modification of the equation of Perkins and Leppert for both gases and liquids. Whitaker [8] proposed a modified version of his own correlation for spheres rather than deriving a new set of coefficients and exponents for cylinders. Fand and Keswani [9] recently presented an empirical correlation for air only, based on the data of Hilpert [4], Collis and Williams [10], and King [11] only. Gnielinski [12] proposed correlation of the data for plates, cylinders, and spheres at moderate Re by taking the square root of the square of the correlations for the laminar and turbulent regimes; however, he did not have access to the very recent data which greatly extend the upward range of Re for cylinders.

The objective of this paper is to resolve, insofar as possible, the discrepancies and complications enumerated in the first paragraph, and to construct a simpler, more accurate, and more general correlation than those mentioned in the second and third paragraph of this

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Table 1 Power-dependence provided by equation (2)

Pr	0	10^{-2}	10^{-1}	0.7	1.0	10	10^{2}	10^{3}	∞
Power	0.500	0.486	0.452	0.401	0.391	0.351	0.337	0.334	0.3333

section. This is not a review. In view of the comprehensive reviews mentioned in the foregoing and the other recent ones by Zhukauskas [13] and Morgan [14], reference will be made only to those investigations directly relevant to the development of the correlations.

The recent data of Achenbach [15] and Zdanavichyus, et al. [16] have helped greatly to resolve the previously controversial behavior at high Re, and the recent papers by Morgan [14] and Katinas, et al. [17] have helped to categorize the various investigations in terms of blockage and free-stream turbulence.

The development herein of an overall correlating equation starts with consideration of the laminar boundary layer regime. The creeping-flow regime and the regime dominated by the wake are next examined. These regimes do not overlap with the laminar boundary layer regime, and interpolation is necessary. Finally, secondary effects are considered, the various sets of data are interpreted briefly in terms of the correlation, and conclusions are drawn.

Development of Correlating Equations

Laminar Boundary Layer Regime. The theoretical solutions for the laminar boundary layer regime proved to be invaluable in constructing correlating equations for forced convection to a flat plate [18] and for free convection to a cylinder [19]. Unfortunately comparable solutions are incomplete for forced convection to a cylinder.

Squire [20], Eckert [21], and Levy [22] all computed theoretical values of the Nusselt number at the forward stagnation point as a function of the Prandtl number. The values of Squire and Eckert are in reasonable agreement but those of Levy are somewhat lower. The values of Squire and of Eckert were, therefore, utilized to derive an empirical expression in the form of the general correlating equation of Churchill and Usagi [23]. This model equates the nth power of the dependent variable (here Nu) to the sum of the nth powers of the limiting solutions for large and small values of the independent variable (here Pr). The empirical exponent n is then evaluated from experimental data or theoretical points for intermediate values of the independent variable. In this application the best value of n appears to be -4.0 resulting in the expression

$$\mathrm{Nu}_s = 1.276\,\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}/[1 + (0.412/\mathrm{Pr})^{2/3}]^{1/4} \qquad (1)$$

It may be noted that the exponent of ¼ (corresponding to $n = -4$) on the term in brackets is identical to that derived by Churchill and Ozoe [24, 25] for forced convection to both uniformly heated and isothermal plates in the laminar boundary layer regime. However, the "central values" of Pr for the flat plate (the values of Pr for which the limiting solutions intersect) were much lower than 0.412. For free convection the central values as well as the exponent were found by Churchill and Churchill [26] to be essentially the same for all geometries. The difference in the central values of Pr for forced convection undoubtedly arises from the basic difference in the forced-flow pattern over the cylinder and along the flat plate. The value of 0.412 is, in any event, quite uncertain since it is the ratio of the coefficients of the limiting solutions for $\mathrm{Pr} \to 0$ and ∞, raised to the sixth power.

Pending the derivation of sufficient, precise values for the effect of Pr on the mean rate of heat transfer over the entire cylinder, the same dependence as in equation (1) will be postulated, yielding

$$\overline{\mathrm{Nu}} = A\,\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}/[1 + (0.4/\mathrm{Pr})^{2/3}]^{1/4} \qquad (2)$$

The approximate, theoretical calculations of Masliyah and Epstein [27] for Re = 1 and Pr from 0.7 to 40,000 indicate a value of 0.62 and the two, extrapolated, theoretical values of Jain and Goel [28] a value of 0.64 for A. An arithmetic plot (not shown) of $\overline{\mathrm{Nu}}$ versus $\phi = \mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}/[1 + (0.4/\mathrm{Pr})^{2/3}]^{1/4}$ for experimental and theoretical values covering a wide range of Re and Pr indicates that equation (2) with $A = 0.62$ provides an excellent representation for $40 < \mathrm{Re} < 10{,}000$ corresponding to $5 < \phi < 80$ for Pr = 0.7. This agreement is illustrated in logarithmic form in Figs. 1 and 2. Equation (2) does not hold for $\mathrm{Re} < 5$ owing to thickening of the boundary layer as discussed in the next section and does not hold for $\mathrm{Re} > 10{,}000$ owing to the development of a significant contribution from the region of separation at the rear of the cylinder, as discussed later.

The equivalent power dependence on Pr in equation (2), defined as $d(\ln\{\overline{\mathrm{Nu}}\})/d(\ln\{\mathrm{Pr}\})$, varies from ½ to ⅓ as indicated in Table 1, thus providing a rationalization for the different powers derived in prior correlations of experimental data for different ranges of Pr.

Creeping-Flow Regime (Pe < 0.2). A number of theoretical expressions have been derived for $\mathrm{Pe} \to 0$, based on potential flow. The most successful appears to be that of Nakai and Okazaki [29] which can be written as

$$\overline{\mathrm{Nu}} = 1/(0.8237 - \ln\{\mathrm{Pe}^{1/2}\}), \qquad \mathrm{Pe} \le 0.2 \qquad (3)$$

Equation (3) agrees well with the computed values of Dennis, et al. [30] and with the experimental data for various fluids, as indicated by the dashed lines in Fig. 1 for Pr = 0.001, 0.7, and 1000, and discussed subsequently. The values of $\overline{\mathrm{Nu}}$ computed from equation (2) approach 0 as $\mathrm{Pe} \to 0$, as would be expected for pure conduction from an infinitely long cylinder to surroundings of infinite extent. The finite values observed experimentally for $\mathrm{Re} \to 0$ are presumably due to free convection, end effects and finite surroundings.

Nomenclature

A = dimensionless coefficient
D = cylinder diameter
D_T = tunnel diameter
$\mathcal{D}$ = diffusivity
g = gravitational acceleration
h = heat transfer coefficient
k = thermal conductivity
k_c = mass transfer coefficient
L = cylinder length
n = arbitrary exponent in Churchill-Usagi equation
Nu = Nusselt number = hD/k
$\overline{\mathrm{Nu}}$ = mean Nusselt number
$\overline{\mathrm{Nu}}_0$ = mean Nusselt number for $\mathrm{Re} \to 0$
Nu_r = Nusselt number in region of wake
Nu_s = Nusselt number at forward stagnation point
$\overline{\mathrm{Nu}}_\infty$ = mean Nusselt number for $\mathrm{Re} \to \infty$
Pe = Peclet number = RePr = DU_b/α
Pr = Prandtl number = ν/α
Ra = Rayleigh number = $g\beta\Delta TD^3/\nu\alpha$
Re = Reynolds number = DU_b/ν
Sc = Schmidt number = $\nu/\mathcal{D}$
Sh = Sherwood number = $k_cD/\mathcal{D}$
T = temperature
Tu = intensity of free-stream turbulence = $\sqrt{(U_b')^2}/U_b$
U_b = free-stream velocity
U_b' = fluctuation in free-stream velocity
α = thermal diffusivity
β = volumetric coefficient of expansion with temperature
$\Delta T = T_b - T_w$
μ = viscosity
ν = kinematic viscosity
$\phi = \mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}/[1 + (0.4/\mathrm{Pr})^{2/3}]^{1/4}$

Subscripts

b = free stream
f = film, at $(T_w + T_b)/2$
w = wall

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Table 2 Power-dependence provided by equation (8) for Pr = 0.7

Re	0	10^{-2}	10^{-1}	1	10	10^2	10^3	10^4	10^5	10^6	10^7	∞
Power	0	0.069	0.169	0.308	0.419	0.471	0.505	0.552	0.671	0.843	0.951	1.000

Intermediate Regime (Pe > 0.2, Re < 10,000). A considerable gap exists between the range of applicability of equations (2) and (3). This behavior can be approximated, as suggested by Tsubouchi and Masuda [7] and others, by adding a constant term, $\overline{\mathrm{Nu}}_0$, to the right-hand side of equation (2). A value of 0.3 resulting in

$$\overline{\mathrm{Nu}} = 0.3 + 0.62\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}/[1 + (0.4/\mathrm{Pr})^{2/3}]^{1/4} \quad (4)$$

represents the experimental data and computed values very well for Pe > 0.2 and Re < 10,000 as indicated in Figs. 1 and 2. A slightly lower value of $\overline{\mathrm{Nu}}_0$ would give a better prediction for Pe < 0.2 but at the expense of a poorer prediction for Pe > 0.2.

The theoretical solution of King [11] for potential flow actually yielded a value of $1/\pi = 0.318$ for Nu_0 but his choice of boundary conditions has been called improper by Hill and Sleicher [31].

Collis and Williams [10] assert that an expression of the form of equation (4) is unsatisfactory because it cannot reproduce the discrete change in slope which they observe with the onset of eddy-shedding (Re = 44 for Pr = 0.7). However, such a discrete transition in $\overline{\mathrm{Nu}}$ at Re = 44 ($\phi = 5.16$ for Pr = 0.7) is not apparent in Fig. 1 and can be considered negligible for practical purposes. A possible explanation is the derivation by Virk [32] of

$$\mathrm{Nu}_r = 2\sqrt{\mathrm{Pe}/5\pi} \quad (5)$$

for the rear half of the cylinder in the eddy-shedding regime of $40 < \mathrm{Re} < 2 \times 10^5$. The dependence on Re is thus the same as for the boundary layer, although the indicated dependence on Pr differs.

Completely Turbulent Regime. The behavior for Re > 10,000 has been uncertain for some time owing to discrepancies between the various sets of experimental data. Clark [33] predicted a linear dependence of Nu on Re in the region of the wake. Both van der Hegge Zijnen [34] and Douglas and Churchill [1] utilized such a linear dependence in their correlating equations. Richardson [6, 35] asserted that the rate of heat transfer in this region should be proportional to $\mathrm{Re}^{2/3}$. This dependence and also $\mathrm{Re}^{0.8}$ have been utilized in a number of correlating equations. The precise data of Hilpert [4] which extend to Re = 233,000, indicate a linear dependence but have been subject to some controversy since they generally fall below the data of other investigators. The recent precise data of Zdanavichyus, et al. [16] and of Achenbach [15] for air which extend to $\mathrm{Re} = 1.1 \times 10^6$ and $\mathrm{Re} = 4.2 \times 10^6$, respectively, now provide a sounder basis for interpretation of the behavior at high Re. (Despite their relative precision, the data of Achenbach for uniform wall temperature and $T_u \cong 0.0045$ fall above those of Zdanavichyus, et al., for uniform heat flux and $T_u \cong 0.027$, even though the opposite effect would be expected.) The evaluation of the role of blockage and free-stream turbulence by Morgan [14] and others explains most of the discrepancies between the other sets of data, at least qualitatively.

The data of both Zdanavichyus, et al., and Achenbach indicate that $\overline{\mathrm{Nu}}$ does not increase uniformly with Re but shows an upswing at about $\mathrm{Re} = 10^5$, a flattening out at about $\mathrm{Re} = 4 \times 10^5$ and a resumed upswing approaching linearity with Re above $\mathrm{Re} = 2 \times 10^6$. These transitions are associated with the downstream relocation of the point of separation and the transition from a laminar to a turbulent boundary layer. Reference to these papers is suggested for detailed discussions of the behavior of the flow field. The changes in $\overline{\mathrm{Nu}}$ are surprisingly mild, considering the abrupt and significant changes in the flow pattern.

The following asymptotic expression for very large Re can be derived from the data of Achenbach

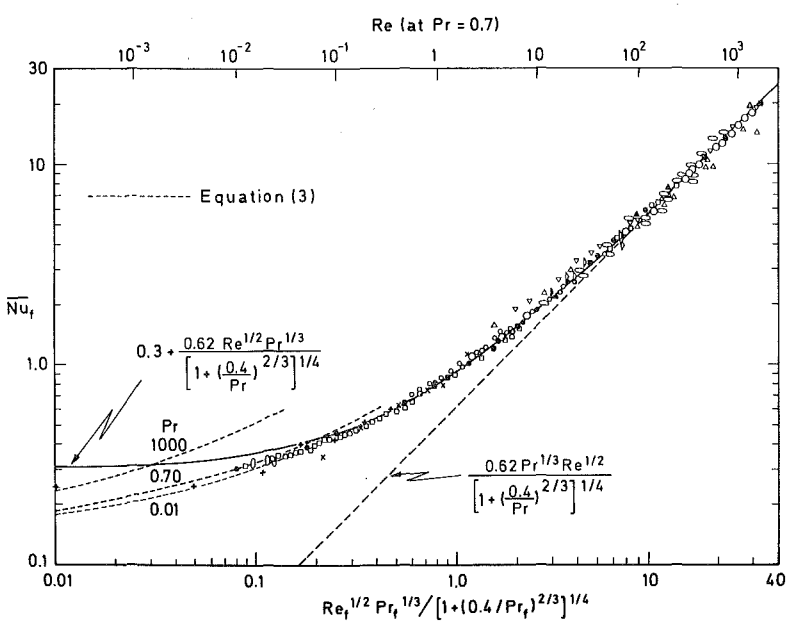

Fig. 1 Low Reynolds number regime—Legend: ● Dennis, et al. [30] (computed); ■ Masliyah and Epstein [27] (computed); ▲ Jain and Goel [28] (computed); ○ Hilpert [4] (air); ○ Tsubouchi and Masuda [7] (air); □ Collis and Williams [10] (air); ○ Gebhart and Pera [36] (air); ▽ Krall and Eckert [37] (air (uniform heat flux)); ⊂⊃ Tsubouchi and Masuda [7], Davis [42] (water, oils (Pr = 4-600)); x Sajben [31] (mercury (Pr = 0.0225)); + Beckers, et al. [43] (paraffin oil (Pr = 1000)); ▷ Ishiguro, et al. [40] (sodium (Pr = 0.0058 and 0.0073); △ Dobry and Finn [46] (mass transfer (Sc = 1100 and 1400))

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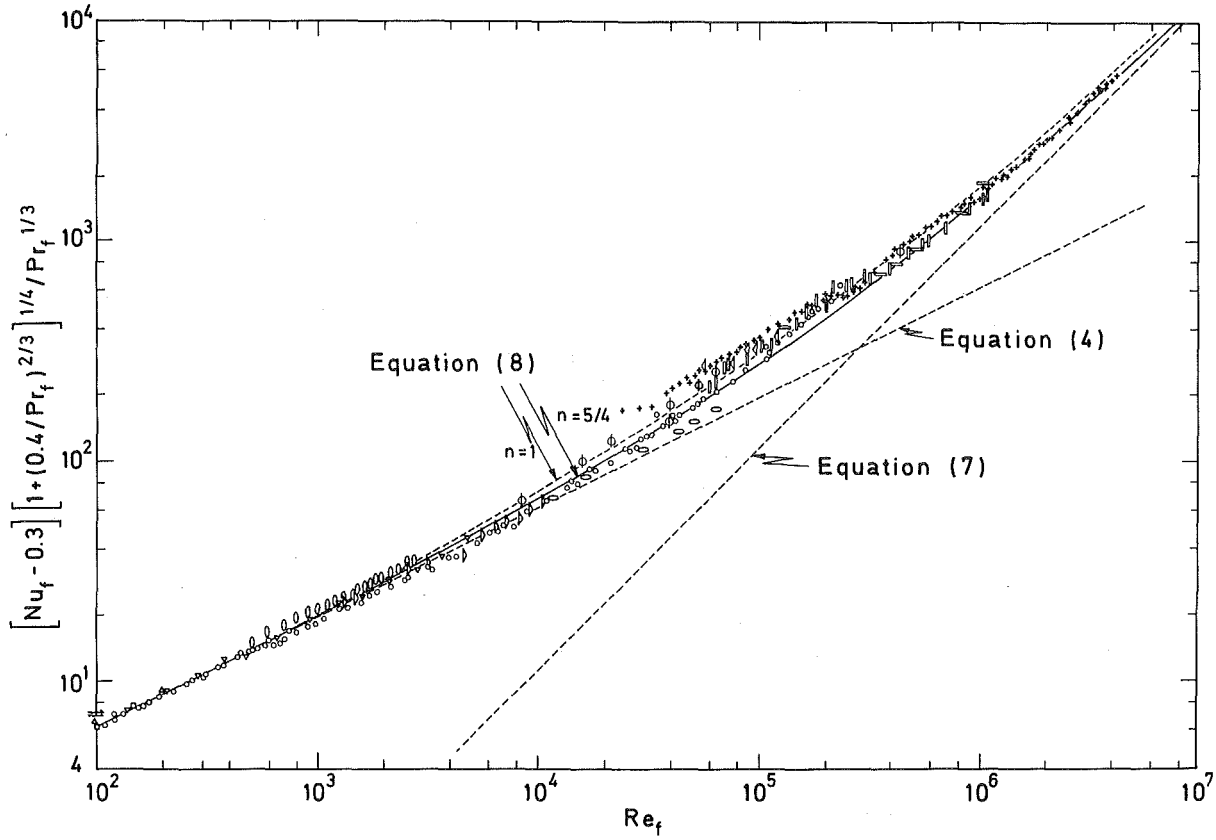

Fig. 2 High Reynolds number regime—Legend: ○ Hilpert [4] (air); □ Collis and Williams [10] (air); + Achenbach [15] (air); ▯ Zdanavichyus, et al. [16] (air); ▽ Krall and Eckert [37] (air (uniform heat flux)); ϕ Schmidt and Wenner [38] (air); ◁ Lewis [39] (air); ▷ Ishiguro, et al. [40] (sodium (Pr = 0.0058 and 0.0073)); ▭ Zdanavichyus, et al. [16] (water); ⦶ Fand and Keswani [44] (water (smoothed values)); ⬭ Fand [45] (water)

$$\overline{Nu}_\infty = 0.00091\ Re \tag{6}$$

Postulating the same dependence on Pr as in the laminar-boundary-layer regime converts equation (6) to

$$\overline{Nu}_\infty = 0.001168\ RePr^{1/3}/[1 + (0.4/Pr)^{2/3}]^{1/4} \tag{7}$$

It should be emphasized that equations (6) and (7) are the apparent asymptotes for the Achenbach data, not correlations for them.

Overall Correlation, $\phi > 0.4$. Combining equations (4) and (7) in the form suggested by Churchill and Usagi [23], as discussed in the foregoing, results in the test expression

$$(\overline{Nu} - 0.3)^n = \left(\frac{0.62Re^{1/2}Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}}\right)^n + \left(\frac{0.001168RePr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}}\right)^n \tag{8}$$

A value of 5/4 for the exponent, hence

$$\overline{Nu} = 0.3 + \frac{0.62Re^{1/2}Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}}\left[1 + \left(\frac{Re}{282000}\right)^{5/8}\right]^{4/5} \tag{9}$$

appears to provide a lower bound for RePr > 0.4 and a reasonable approximation for all Re and Pr, with a few exceptions to be discussed in the next section.

Equation (9) differs from prior correlating equations for forced convection to cylinders in that it provides a varying power for Re and Pr. The power of Pr was previously considered. The effective power of Re in equation (9), defined as $[\partial(\ln\{\overline{Nu}\})/\partial(\ln\{Re\})]_{Pr}$ depends on Pr. The values in Table 2, which were computed for Pr = 0.7, range from 0 to 1.0, and are in general agreement with the values determined by the correlations for narrow ranges of Re by McAdams [2], Morgan [14], and others. The values in Table 1 for the effective power of Pr hold for equations (4) and (9) as well as (2) as long as $\overline{Nu} \gg Nu_0$.

Comparison of Correlating Equations With Experimental Data

The effectiveness and limitations of the correlating equations (3), (4), and (9) are indicated in Figs. 1 and 2 by comparison with a broad range of experimental data. The properties and velocity were converted to the form of Figs. 1 and 2 insofar as possible. The sets of data which are included are those which do not scatter excessively, and which Morgan [14] and others have classified as relatively free from the effects of temperature-difference, free-stream turbulence, aspect ratio and tunnel-blockage. The many sets of early data (see, for example, Douglas and Churchill [1], Gnielinski [12], or Morgan [14] for listings) which are not included, generally fall significantly above equation (9) due to one or more of the foregoing effects. Corrections for these secondary effects are considered in the following section.

The various sets of data for air [4, 7, 10, 15, 16, 36–39] are seen to be in good agreement with equation (9) for Pe > 0.2, corresponding to Re = 0.285 and ϕ = 0.416, with the following exceptions. The data of Krall and Eckert [37] fall somewhat above equation (9) for their lower range of Re, presumably due to uniform heating. In the range 40000 < Re < 400000 all of the data [4, 15, 16, 38, 39] fall decisively above equation (9). This thermal behavior is associated with the downstream transition in the point of separation of the wake. A somewhat better representation for the data in this range of Re is obtained by taking n = 1 instead of 5/4 in equation (8) leading to

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$$\overline{\mathrm{Nu}} = 0.3 + \frac{0.62\mathrm{Re}^{1/2}\mathrm{Pr}^{1/3}}{[1 + (0.4/\mathrm{Pr})^{2/3}]^{1/4}}\left[1 + \left(\frac{\mathrm{Re}}{282000}\right)^{1/2}\right] \quad (10)$$

Even so, some of the data of Achenbach [15] and of Lewis [39] fall above equation (10). In the range $1000 < \mathrm{Re} < 10{,}000$ the precise data of Hilpert [4] fall below equation (9). This latter discrepancy, which has been noted by many previous investigators, may be due to some experimental anomaly, as suggested by Morgan [14].

Reliable data to test the dependence on Pr are somewhat limited. The only reliable data for low Pr appear to be those of Sajben [34] for mercury (Pr = 0.0225) and those of Ishiguro, et al. [40] for sodium (Pr = 0.0058 and 0.0073). Representative values of Sajben read from a graph, are included in Fig. 1. They are in general agreement with equation (4) for Pe > 0.2, corresponding to $\phi = 0.50$, but fall below both equations (4) and (3) for lower values. The measurements of Ishiguro, et al., are for a nearly uniform heat flux density. The values of $\overline{\mathrm{Nu}}$ based on an integrated-mean temperature difference appear to follow equation (4) even to Re = 10000, and hence fall below equation (9) for $\mathrm{Re} > 10^3$. Based on an integrated-mean heat transfer coefficient they fall above equation (9). In any event they appear to invalidate, for this range of Re, the postulate of potential flow, upon which the theoretical solution of Grosh and Cess [41] is based.

The data of Tsubouchi and Masuda [7] and Davis [42] for various fluids with $5 < \mathrm{Pr} < 600$ scatter randomly about equation (9). The data of Beckers, et al. [43] for paraffin oil (Pr = 1000) agree with equation (4) down to Pe = 0.2($\phi = 0.14$) but fall far below the prediction of equation (3) for lower Pe. Their data for other fluids scatter widely and were omitted. The data of van der Hegge Zijnen [34] are given graphically in combined forms which preclude their inclusion in Figs. 1 and 2. The data of Gebhart and Pera [36] for oils and small wires were recognized by the investigators as anomalously high, apparently due to end effects. The data of Zdanavichyus, et al. [16] for water follow equation (9) closely and those of Fand and Keswani [44] are only slightly higher. The earlier data of Fand [45] for water fall decisively below the correlation, and almost follow the extension of equation (4), suggesting a possible delay in transition of the point of separation in these experiments.

Although the correlation is presumed to hold for mass transfer with $\overline{\mathrm{Nu}}$ replaced by $\overline{\mathrm{Sh}}$ and Pr by Sc, no data meeting the foregoing criteria were found. However, the data of Dobry and Finn [46] for Sc = 1100 and 1400, although somewhat scattered, are seen to be in general agreement with equation (9).

Secondary Effects

Physical Property Variation With Temperature. The effect of large ΔT is not well resolved and is opposite in direction for liquids and gases and for heating and cooling. Douglas and Churchill [1] were reasonably successful in handling the effect of property variation for gases and Tsubouchi and Masuda [7] for liquids simply by calculating the properties (including ρ) at the film temperature (the arithmetic average of the surface and free-stream temperatures). Collis and Williams [10] assert that for gases $\overline{\mathrm{Nu}}$ should additionally be multiplied by $(T_b/T_f)^{0.17}$. Perkins and Leppert [5] and Whitaker [8] propose the use of free-stream properties and the multiplication of $\overline{\mathrm{Nu}}$ by $(\mu_w/\mu_b)^{0.25}$ but Tsubouchi and Masuda contend an exponent of 0.14 is better than 0.25. Zhukauskas [13] proposes the use of the free-stream temperature and the multiplication of $\overline{\mathrm{Nu}}$ by $(\mathrm{Pr}_b/\mathrm{Pr}_w)^\alpha$ with $\alpha = 0.25$ for the turbulent regime and 0.17–0.19 for the laminar regime, but notes that $\mathrm{Pr}_b/\mathrm{Pr}_w \cong \mu_b/\mu_w$. Fand and Keswani [44] propose correlating $\mathrm{Nu}_w{}^a\mathrm{Nu}_b{}^{1-a}$ with $\mathrm{Re}_w{}^b\mathrm{Re}_b{}^{1-b}$ and $\mathrm{Pr}_w{}^c\mathrm{Pr}_b{}^{1-c}$ but do not achieve a significantly better fit than with properties at the film temperature. After reviewing the various correlations, Morgan concludes that the dependence on physical properties is still undefined and declares the need for additional experimental work. Wylie [47] and Vaitekunas, et al. [48] have recently derived theoretical solutions for laminar and turbulent boundary layers, respectively, with varying physical properties and this approach may ultimately resolve the dilemma.

In any event, equation (9) can be expected to give reasonable results for moderate temperature differences for both liquids and gases if the properties are evaluated at the film temperature. It can readily be modified in accordance with any of the foregoing schemes for large temperature differences.

Boundary Conditions. Most experiments and calculations have been carried out for essentially a uniform surface temperature. However, Boulos and Pei [49] utilized a uniform heat flux density and concluded that this boundary condition resulted in a 10–20 percent increase in $\overline{\mathrm{Nu}}$ as compared to an isothermal surface. Some of their measured values of $\overline{\mathrm{Nu}}_s$ are below the boundary layer values, casting doubt on the validity of this conclusion. Krall and Eckert [37] concluded that their measurements were 7 percent high for the same reason. These observations are consistent with theoretical calculations for the laminar boundary layer regime on a flat plate which indicate a 37 percent increase in the local values of Nu but a 37 percent increase, 3 percent increase or 3 percent decrease depending on whether $\overline{\mathrm{Nu}}$ is based on the integrated-mean heat transfer coefficient, the integrated-mean temperature-difference or the temperature difference at the midpoint [18]. These results provide the basis for a first-order estimation of a correction factor to equation (9) for a uniform heat flux or other nonisothermal boundary condition

Blockage. Morgan [14] compared a number of the expressions that have been proposed for the effect of the boundaries of the channel in which the test cylinder is located, with the limited experimental data for this effect. He concluded that the velocity in Re should be increased by the factor $1 + 0.385\, D/D_T + 1.356\,(D/D_T)^2$ developed by Vincenti and Graham [50] for small streamlined bodies in closed circular tunnels. For a jet stream from a tunnel, Morgan suggests reducing the velocity in Re by the factor $1 - 0.411\,(D/D_T)^2$ which was developed by Lock [51].

Free-Stream Turbulence. Dyban and Epik [52] proposed the expression 0.8 ReTu/(1500 + ReTu) to represent experimental data for the fractional increase in $\overline{\mathrm{Nu}}$ over the range $2000 < \mathrm{Re} < 80{,}000$ and $0.02 < \mathrm{Tu} < 0.26$. Morgan [14] reviewed the somewhat contradictory data and correlations and developed a graphical correlation for the combined effects of free-stream turbulence and tunnel blockage.

Aspect Ratio. $\overline{\mathrm{Nu}}$ depends on the length of the cylinder owing to three-dimensional disturbances in the velocity field at the ends, and perhaps to longitudinal heat losses. These effects appear to be significant only for very fine wires such as those used in anemometry. Öhman [53] has shown that $\overline{\mathrm{Nu}}_0$ should have a minimum theoretical value of $2/\ln\{2L/D\}$ for a finite cylinder. Gebhart and Pera [36] observed significant effects for large Pr, even for $L/D = 16000$. Morgan [14] proposes multiplication of $\overline{\mathrm{Nu}}$ for hot-wire anemometers by the factor $1 + 7.5(D/L)^{1/2} + 3.5 \times 10^4(D/L)^2$.

Free Convection. Since forced convection implies a temperature difference, free convection must always be present to some degree. From the general correlation of Churchill [54] for assisting, laminar convection it follows that the fractional increase or decrease in $\overline{\mathrm{Nu}}/\overline{\mathrm{Nu}}_0$ due to a small degree of assisting or opposing free convection is approximately $0.2\ \mathrm{Ra}^{3/4}\ [1 + (0.4/\mathrm{Pr})^{2/3}]^{3/4}/\mathrm{Re}^{3/2}\mathrm{Pr}[1 + (0.5/\mathrm{Pr})^{9/16}]^{4/3}$. For very small Re and Ra Nakai and Okazaki [29] show that the absolute increase or decrease in 1/Nu due to a small degree of opposing or assisting free convection is $0.65\ \overline{\mathrm{Nu}}\mathrm{Ra}/\mathrm{Pr}^2\mathrm{Re}^3$.

Summary and Conclusions

1 Equation (9) is proposed as a lower bound for the computed and experimental values of heat transfer by forced convection to a cylinder in cross flow for all Re and Pr such that RePr > 0.2.

2 As a lower bound, equation (9) presumably represents the behavior for low free-stream turbulence, an isothermal surface, negligible blockage, negligible end-effects, a small temperature difference and negligible free convection. A possible exception is in the range of $10^3 < \mathrm{Re} < 10^4$ where the data of Hilpert [4] for air and of Ishiguro, et al. [40] for sodium appear to follow equation (4) rather than equation (9). In the range of $7 \times 10^4 < \mathrm{Re} < 4 \times 10^5$, $\overline{\mathrm{Nu}}$ may be significantly higher than predicted by equation (9) owing to a downstream shift of the point of separation of the laminar boundary layer. Equation (10) can be used as a lower bound and predictor in this range.

3 Equation (4) can be used as an approximation for equation (9)

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for Re < 4000 and all Pr.

4 Equation (3), which is based on the assumption of creeping flow, should provide a better representation than equations (4) or (9) for RePr < 0.2 if free convection and end-effects are negligible. It agrees well with such experimental data for air, but has not been tested critically for a wide range of Pr.

5 Equation (4) appears to provide reasonably good predictions even for RePr < 0.2 and can be modified to provide an even better representation for any single set of data by the proper, arbitrary choice of Nu_0.

6 Statistical methods were not used to evaluate the constants in equations (4) and (9) nor to characterize the scatter since the various sets of data are obviously of widely differing and unknown quality and in some cases were determined with some uncertainty from graphs.

7 Equation (9) has the advantage over prior correlating equations in that it takes into account the recent data of Achenbach [15] and Zdanavichyus, et al. [16] for an extended range of Re.

8 Equation (9) differs from prior correlating equations in that it is based, insofar as possible, on theoretical results, and provides a continuously varying dependence on both Re and Pr.

9 The success of equation (9) is quite remarkable in view of the many strong transitions in the flow field as documented by Collis and Williams [10], Achenbach [16] and others for air. Fortunately, the corresponding transitions in the heat transfer coefficient are quite weak and compensatory with the exception noted previously for $1 \times 10^5 < Re < 4 \times 10^5$.

10 An empirical equation could be constructed which follows these transitions but the behavior is not yet clearly enough defined to justify such a complex construction.

11 Expressions are suggested to estimate the increased or decreased value of $\overline{Nu}$ due to free-stream turbulence, nonisothermal boundary conditions, tunnel blockage, end-effects, large temperature-differences and free convection.

12 Additional data are still needed for the very low and the intermediate and high range of Re and for a wide range of Pr or Sc. In order to be of value these measurements must be of good precision and accuracy and free from the effects enumerated in conclusion 11.

13 Tabulations of experimental and computed values of Nu, Re, and Pr should be included in future publications for the convenience of subsequent workers.

14 Equation (9) and (4) are presumed to be applicable for mass transfer with $\overline{Sh}$ and Sc substituted for $\overline{Nu}$ and Pr, although values to test this generalization are rather limited.

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References

1 Douglas, W. J. M., and Churchill, S. W., "Recorrelation of Data for Convective Heat Transfer Between Gases and Single Cylinders With Large Temperature Differences," *Chemical Engineering Progress Symposium Series*, No. 18, Vol. 52, 1956, pp. 23–28.

2 McAdams, W. H., *Heat Transmission*, Third ed., McGraw-Hill, New York, 1954, p. 267.

3 Fand, R. M., and Keswani, K. K., "Recalculation of Hilpert's Constants," JOURNAL OF HEAT TRANSFER, TRANS. ASME, Series C, Vol. 95, 1973, pp. 224–226.

4 Hilpert, R., "Wärmeagabe von geheizten Drähten and Rohren im Luftstrom," *Forschungsarbeiten auf dem Gebiete des Ingenieurwesens*, Vol. 4, 1933, pp. 215–224.

5 Perkins, H. C., Jr., and Leppert, G., "Forced Convection Heat Transfer From a Uniformly Heated Cylinder," JOURNAL OF HEAT TRANSFER, TRANS. ASME, Series C, Vol. 84, 1962, pp. 257–263.

6 Richardson, P. D., "Heat and Mass Transfer in Turbulent Separated Flows," *Chemical Engineering Science*, Vol. 18, 1963, pp. 149–155.

7 Tsubouchi, T., and Masuda, H., "On the Experimental Formulae of Heat Transfer From Single Cylinders by Forced Convection," *Reports of the Institute of High Speed Mechanics*, Tohoku University, Japan, Vol. 19, 1967-1968, pp. 221–239.

8 Whitaker, S., "Forced Convection Heat Transfer Correlation for Flow in Pipes, Past Flat Plates, Single Cylinders, Single Spheres, and for Flow in Packed Beds and Tube Bundles," *AIChE Journal*, Vol. 18, 1972, pp. 361–371.

9 Fand, R. M., and Keswani, K. K., "A Continuous Correlation Equation for Heat Transfer From Cylinders to Air in Crossflow for Reynolds Numbers From 10^{-2} to 2×10^5," *International Journal of Heat and Mass Transfer*, Vol. 15, 1972, pp. 559–562.

10 Collis, D. C., and Williams, M. J., "Two-dimensional Convection From Heated Wires at Low Reynolds Numbers," *Journal of Fluid Mechanics*, Vol. 6, Part 3, 1959, pp. 357–384.

11 King, L. V., "On the Convection of Heat From Small Cylinders in a Stream of Fluid," *Philosophical Transactions of the Royal Society* (London), Series A, Vol. 214, 1914, pp. 373–432.

12 Gnielinski, V., "Berechnung mittlerer Wärme- und Stoffübergangskoeffizienten an laminar und turbulent überströmten Einzelkörpern mit Hilfe einer einheitlichen Gleichung," *Forschung im Ingenieurwesen*, Vol. 41, 1975, pp. 145–153.

13 Zhukauskas, A., "Heat Transfer From Tubes in Crossflow," *Advances in Heat Transfer*, Vol. 8, Academic Press, New York, 1972, pp. 93–160.

14 Morgan, V. T., "The Overall Convective Heat Transfer From Smooth Circular Cylinders," *Advances in Heat Transfer*, Vol. 11, Academic Press, New York, 1975, pp. 199–264.

15 Achenbach, "Total and Local Heat Transfer From a Smooth Circular Cylinder in Cross-flow at High Reynolds Number," *International Journal of Heat and Mass Transfer*, Vol. 18, 1975, pp. 1387–1396.

16 Zdanavichyus, G. B., Chesna, B. A., Zhyugzhda, I. I., and Zhukauskas, A. A., "Local Heat Transfer in an Air Stream Flowing Laterally to a Circular Cylinder at High Re Numbers," *Intern'l. Chem. Engng.*, Vol. 16, 1976, pp. 121–127.

17 Katinas, V. I., Shvegzhda, S. A., Zhyugzhda, I. I., and Zhukauskas, A. A., "Streamline Flow and Heat Transfer at the Frontal Section of a Circular Cylinder in Turbulent Streams of a Viscous Liquid," *Intern'l. Chem. Engng.*, Vol. 16, 1976, pp. 469–475.

18 Churchill, S. W., "A Comprehensive Correlating Equation for Forced Convection From Flat Plates," *AIChE Journal*, Vol. 21, 1976, pp. 264–268.

19 Churchill, S. W., and Chu, H. H.-S., "Correlating Equations for Laminar and Turbulent Free Convection From a Horizontal Cylinder," *International Journal of Heat and Mass Transfer*, Vol. 18, 1975, pp. 1049–1053.

20 Squire, H. B., in "*Modern Developments in Fluid Dynamics*, S. Goldstein, ed., Clarendon Press, Oxford, England, 1938, Vol. II, p. 631.

21 Eckert, E. R. G., "Die Berechnung des Wärmeübergangs in der Laminaren Grenzschicht umströmter Körper," *VDI Forschungsheft*, No. 416, 1942.

22 Levy, S., "Heat Transfer to Constant-Property Laminar Boundary-Layer Flows With Power-Function Free-Stream Velocity and Wall-Temperature Variation," *J. of the Aero. Scis.*, Vol. 19, 1952, pp. 341–348.

23 Churchill, S. W., and Usagi, R., "A General Expression for the Correlation of Rates of Transfer and Other Phenomena," *AIChE Journal*, Vol. 18, 1972, pp. 1121–1128.

24 Churchill, S. W., and Ozoe, H., "Correlations for Laminar Forced Convection With Uniform Heating in Flow over a Plate and in Developing and Fully Developed Flow in a Tube," JOURNAL OF HEAT TRANSFER TRANS. ASME, Series C, Vol. 95, 1973, pp. 78–84.

25 Churchill, S. W., and Ozoe, H., "Correlation for Laminar Forced Convection in Flow over an Isothermal Tube," JOURNAL OF HEAT TRANS., TRANS. ASME, Series C, Vol. 95, 1973, pp. 416–419.

26 Churchill, S. W., and Churchill, R. U., "A Comprehensive Correlating Equation for Heat and Component Transfer by Free Convection," *AIChE Journal*, Vol. 21, 1975, pp. 604–607.

27 Masliyah, J. H., and Epstein, N., "Heat and Mass Transfer From Elliptical Cylinders in Steady Symmetric Flow," *Ind. and Engng. Chem. Funda.*, Vol. 12, 1973, pp. 317–323.

28 Jain, P. C., and Goel, B. S., "A Numerical Study of Unsteady Laminar Forced Convection From a Circular Cylinder," JOURNAL OF HEAT TRANSFER, TRANS. ASME, Series C, Vol. 98, 1976, pp. 303–307.

29 Nakai, S., and Okazaki, T., "Heat Transfer From a Horizontal Circular Wire at Small Reynolds and Grashof Numbers—I Pure Convection," *International Journal of Heat and Mass Transfer*, Vol. 18, 1975, pp. 387–396.

30 Dennis, S. C. R., Hudson, J. D., and Smith, N., "Steady Laminar Forced Convection From a Circular Cylinder at Low Reynolds Numbers," *Physics of Fluids*, Vol. 11, 1968, pp. 933–940.

31 Hill, J. C., and Sleicher, C. A., Jr., "Convective Heat Transfer From Small Cylinders to Mercury," *International Journal of Heat and Mass Transfer*, Vol. 12, 1969, pp. 1595–1604.

32 Virk, P. S., "Heat Transfer From the Rear of a Cylinder in Transverse Flow," JOURNAL OF HEAT TRANSFER, TRANS. ASME, Series C, Vol. 92, 1970, pp. 206–207.

33 Clark, L. G., private communication (see reference [1]).

34 van der Hegge Zijnen, "Modified Correlation Formulae for the Heat Transfers by Natural and by Forced Convection From Horizontal Cylinders, *Applied Sci. Res.*, Series A, Vol. 6, 1956, pp. 129–140.

35 Richardson, P. D., "On Hilpert's Measurements of Heat Transfer From Cylinders Transverse to an Air Stream," JOURNAL OF HEAT TRANSFER, TRANS. ASME, Series C, Vol. 85, 1963, pp. 283–284.

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36 Gebhart, B., and Pera, L., "Mixed Convection From Long Horizontal Cylinders," *Journal of Fluid Mechanics*, Vol. 45, 1970, pp. 49–64.

37 Krall, K. M., and Eckert, E. R. G., "Local Heat Transfer Around a Cylinder at Low Reynolds Number," JOURNAL OF HEAT TRANS., TRANS. ASME, Series C, Vol. 95, 1973, pp. 273–275.

38 Schmidt, E., and Wenner, K., "Wärmeabgabe über den Umfang eines angeblasenen geheizten Zylinders.," *Forschung im Ingenieurwesen*, Vol. 12, 1941, pp. 65–73.

39 Lewis, J. S., "Heat Transfer for Predictions From Mass Transfer Measurements Around a Single Cylinder in Cross Flow," *International Journal of Heat and Mass Transfer*, Vol. 14, 1971, pp. 325–329.

40 Ishiguro, R., Kumada, T., Sugujama, and Ikezaki, E., "Experimental Study of Heat Transfer Around a Circular Cylinder in a Liquid-Sodium Crossflow," *Intern'l. Chem. Engng.*, Vol. 16, 1976, pp. 249–252.

41 Grosh, R. J., and Cess, R. D., "Heat Transfer to Fluids With Low Prandtl Numbers for Flow Across Plates and Cylinders of Various Cross Section," TRANS. ASME, Vol. 80, 1958, pp. 667–676.

42 Davis, A. H., "Convective Cooling of Wires in Streams of Viscous Liquids," *Philosophical Magazine*, Vol. 47, 1924, pp. 1057–1092.

43 Beckers, H. L., ter Haar, L. W., Tjoan, L. T., Merk, H. J., Prins, J. A., and Schenk, J., "Heat Transfer at Very Low Grashof and Reynolds Numbers," *Appl. Sci. Res.*, Series A, Vol. 6, 1956, pp. 82–84.

44 Fand, R. M., and Keswani, K. K., "The Influence of Property Variation on Forced Convection Heat Transfer to Liquids," *International Journal of Heat and Mass Transfer*, Vol. 15, 1972, pp. 1515–1536.

45 Fand, R. M., "Heat Transfer by Forced Convection From a Cylinder to Water in Crossflow," *International Journal of Heat and Mass Transfer*, Vol. 8, 1965, pp. 995–1010.

46 Dobry, R., and Finn, R. K., "Mass Transfer to a Cylinder at Low Reynolds Numbers," *Ind. and Engng. Chem.*, Vol. 48, 1956, pp. 1540–1543.

47 Wylie, R. G., "The Transfer Coefficients of a Laminar Boundary Layer With Variable Fluid Properties," *The Chem. Engng. J.*, Vol. 6, 1973, pp. 1–14.

48 Vaitekunas, P. P., Schlanshyyanskas, A. A., and Zhukauskas, A. A., "Influence of the Variable Physical Properties of a Gas on Drag and Heat Transfer at a Plate With Developed Turbulent Flow," *Intern'l. Chem. Engng.*, Vol. 16, 1976, pp. 403–411.

49 Boulos, M. I., and Pei, D. C. T., "Dynamics of Heat Transfer From Cylinders in a Turbulent Air Stream," *International Journal of Heat and Mass Transfer*, Vol. 17, 1974, pp. 767–783.

50 Vincenti, W. G., and Graham, D. J., "The Effect of Wall Interference Upon the Aerodynamic Characteristics of an Airfoil Spanning a Closed-Throat Circular Wind Tunnel," National Advisory Committee Aeronautics Technical Report No. 849, Washington, D. C., 1946.

51 Lock, C. N., "The Interference of a Wind Tunnel on a Symmetrical Body," Aeronautical Research Council, Report, Memo No. 1275, 1929, No. 1451, 1931.

52 Dyban, E. P., and Epik, E. Ya., "Heat Transfer in Laminar Boundary Layer With Increased Turbulence of Outer Flow," *Heat Transfer-Soviet Research*, Vol. 6, 1974, pp. 159–162.

53 Öhman, G. A., "A Note on Experimental Determination of Convective Heat Transfer From Wires at Extremely Small Reynolds and Grashof Numbers," *Acta Acad. Abo*, Series B, Vol. 30, 1970, pp. 1–7.

54 Churchill, S. W., "A Comprehensive Correlating Equation for Laminar, Assisting, Forced and Free Convection," *AIChE Journal*, Vol. 23, 1977, pp. 10–16.

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